🎵 Tunely — Complete Technical Breakdown & Master Interview Guide

Categorized Tool Deep-Dives, Source Code Implementations, Performance Case Studies, and Interview Answers

> 💡 **Complete Categorized Reference**
> Every language, framework, API, database schema, CSS technique, algorithm, and optimization in Tunely is broken down into its own dedicated section below for easy navigation and study.

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1. 🎤 30-Second Elevator Pitch

Use this script when asked: *"Can you tell me about the Tunely project listed on your resume?"*

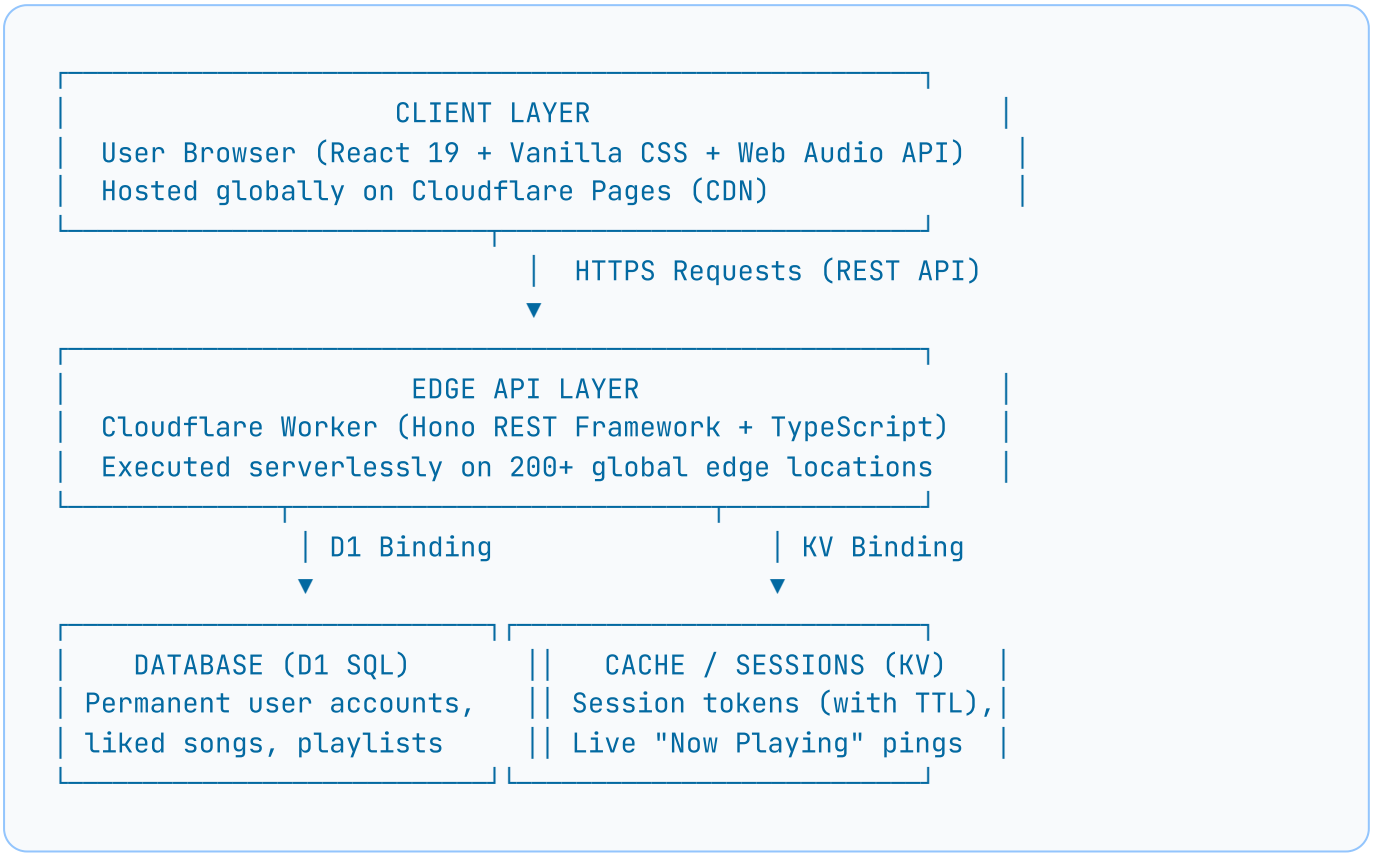
🗣️ Word-for-Word Script:

"Tunely is a full-stack, real-time music streaming web application built with React 19, TypeScript, and a serverless edge backend hosted on Cloudflare Workers. It streams high-quality audio up to 320 kbps, features real-time line-by-line synchronized lyrics display, supports cross-device library synchronization, and includes a live Admin monitoring dashboard.

On the frontend, I used React 19, Vite, Framer Motion, and a custom Vanilla CSS design system with Glassmorphism UI tokens. For audio, I integrated the HTML5 Audio API, a custom 3-band parametric sound equalizer using the Web Audio API, and the OS Media Session API for lock screen playback controls.

On the backend, I built a zero-cold-start edge API using the Hono framework on Cloudflare Workers, Cloudflare D1 (SQLite database) for relational user storage, and Cloudflare KV for session authorization tokens and real-time active user tracking."

2. 🗺️ System Architecture & End-to-End Data Flow



Trace of Playing a Song:

1. **User Click**: Handler `playTrack(track)` is invoked inside `AudioContext.jsx`.
2. **State Mutation**: Updates `currentTrack`, sets `isPlaying = true`, and sets `audioRef.current.src = track.streamUrl`.
3. **HTML5 Audio Streaming**: `audioRef.current.play()` begins streaming audio from CDN.
4. **Media Session API Integration**: Updates lock screen / Android notification bar with song title, artist, and artwork.
5. **Async Lyrics Fetch**: A side-effect (`useEffect`) fires an async request to `/api/songs/{id}/lyrics` via the Worker.
6. **Live Activity Heartbeat**: Every 15 seconds, a heartbeat ping posts to `/api/user/activity`, writing the user's status to Cloudflare KV.

3. ⚛ React 19 & Frontend Core Tech

React 19

UI Library

Why we used it: Declarative component-based structure, efficient virtual DOM diffing algorithm, and functional state management using Hooks.

Where used: Entire frontend UI (Sidebar, PlayerBar, MainContent, SongRow, LyricsPanel, LibraryView, AdminPanel).

React Router DOM v6

Client Routing

Why we used it: Enables Single Page Application (SPA) navigation without triggering full page reloads.

Routes configured: `/home`, `/search`, `/library`, `/playlist/:id`, `/album/:id`, `/custom/:id`.

Framer Motion (v12)

Animation Engine

Why we used it: Provides hardware-accelerated fluid spring animations for page route transitions, hero banner scaling, and staggered card entrances.

Lucide React

Vector SVG Icons

Why we used it: Over 1,000 clean, scalable vector SVG icon components (`Play`, `Pause`, `Heart`, `Search`, `Library`, `Volume2`, `Shuffle`, `Repeat`).

4. 🧠 State & Context API Architecture

To eliminate prop-drilling across 20+ component tiers, global audio playback and authentication state are managed using React Context.

```
// AudioContext State Structure (src/context/AudioContext.jsx)
const AudioContextValue = {
  currentTrack,     // Currently active song object metadata
  isPlaying,        // Boolean flag (true / false)
  currentTime,      // Current playback progress (seconds)
  duration,         // Total track duration (seconds)
  volume,           // Audio volume float (0.0 to 1.0)
  likedSongs,       // Array of liked song objects & IDs
  queue,            // Upcoming auto-play queue list
  playTrack,        // Method to switch active track
  togglePlay,       // Method to pause/resume playback
  toggleLikeTrack   // Method to add/remove from liked songs
};
```

Custom Hooks:

- `useAudio()`: Hook providing components access to playback state and control handlers.
- `useAuth()`: Hook managing user session token, profile data, and login/logout methods.

5. 🎨 Design System & CSS Styling

Vanilla CSS & Design Tokens

Styling System

Built using CSS Custom Properties (variables) defined in `src/index.css` for global dark-mode theme colors (`--primary`, `--bg-dark`, `--panel-bg`) and typography scales.

Glassmorphism UI

Visual Aesthetic

Achieved using `backdrop-filter: blur(32px)`, semi-transparent background colors (`rgba(22, 27, 34, 0.75)`), and subtle 1px white borders (`rgba(255, 255, 255, 0.12)`) for a sleek, premium interface.

Google Fonts Typography Suite

Typography

- **Plus Jakarta Sans**: Main body text & UI labels.
- **Space Grotesk**: Techy display section headings.
- **Outfit**: Secondary title headings.
- **JetBrains Mono**: Timers (`3:45 / 4:20`) and bitrate badges.
- **EB Garamond**: Classic brand logo title font.

6. 🎧 Audio Engineering & Web Audio EQ

1. HTML5 Audio API vs Web Audio API

- **HTML5 Audio API (`new Audio()`)**: Used for streaming, buffering, duration tracking, and listening to media events (`play`, `pause`, `timeupdate`, `ended`).
- **Web Audio API (3-Band Equalizer)**:

```
// 3-Band Parametric EQ Implementation:
const audioCtx = new (window.AudioContext || window.webkitAudioContext)();
const source = audioCtx.createMediaElementSource(audioRef.current);

// Sub-Bass Filter (Low Shelf @ 80 Hz, +1.5 dB)
const bassFilter = audioCtx.createBiquadFilter();
bassFilter.type = 'lowshelf';
bassFilter.frequency.value = 80;
bassFilter.gain.value = 1.5;

// Presence Filter (Peaking @ 3000 Hz, +1.0 dB)
const presenceFilter = audioCtx.createBiquadFilter();
presenceFilter.type = 'peaking';
presenceFilter.frequency.value = 3000;
presenceFilter.gain.value = 1.0;

// Air Filter (High Shelf @ 15000 Hz, +1.5 dB)
const airFilter = audioCtx.createBiquadFilter();
airFilter.type = 'highshelf';
airFilter.frequency.value = 15000;
airFilter.gain.value = 1.5;

// Connect audio nodes in series: source → bass → presence → air → spe
source.connect(bassFilter);
bassFilter.connect(presenceFilter);
presenceFilter.connect(airFilter);
airFilter.connect(audioCtx.destination);
```

2. Media Session API Integration

Provides native OS lock screen controls on Android, iOS, and desktop operating systems:

```
if ('mediaSession' in navigator) {
  navigator.mediaSession.metadata = new MediaMetadata({
    title: track.name,
    artist: track.artists.primary[0]?.name,
    album: track.album.name,
    artwork: [{ src: track.image[2].url, sizes: '500x500', type: 'image/jpeg'
  });

  navigator.mediaSession.setActionHandler('play', () => togglePlay());
  navigator.mediaSession.setActionHandler('pause', () => togglePlay());
  navigator.mediaSession.setActionHandler('nexttrack', () => handleNext());
}
```

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7. 🎤 Synced LRC Lyrics Engine Code

Parses LRC timestamped strings into a synchronized lyrics timeline:

```
// 1. LRC Regex Parsing Function
const parseLyrics = (lrcString) => {
  if (!lrcString) return [];
  const lines = lrcString.split('\n');
  const result = [];
  const regex = /^\[(\d{2}):(\d{2}\.\d{2})\](.*)/;

  for (const line of lines) {
    const match = line.match(regex);
    if (match) {
      const minutes = parseFloat(match[1]);
      const seconds = parseFloat(match[2]);
      const totalTime = minutes * 60 + seconds;
      const text = match[3].trim();
      if (text) result.push({ time: totalTime, text });
    }
  }
  return result;
};

// 2. Active Line Synchronization Logic
const activeIndex = useMemo(() => {
  return parsedLines.findIndex((line, i) => {
    const nextLine = parsedLines[i + 1];
    return currentTime >= line.time && (!nextLine || currentTime < nextLine.t
  });
}, [parsedLines, currentTime]);

// 3. Auto-scroll active line into view
useEffect(() => {
  if (activeLineRef.current) {
    activeLineRef.current.scrollIntoView({ behavior: 'smooth', block: 'center
  }
}, [activeIndex]);
```

8. Cloudflare Workers & Hono API Framework

- **Cloudflare Workers**: Serverless runtime executing code on Cloudflare's global edge network in V8 isolates with **< 5ms startup latency**.
- **Hono Framework**: Lightweight router for Workers with full TypeScript type safety.

```
// Hono Worker Router Example (jiosaavn-api/src/server.ts)
import { Hono } from 'hono';

const app = new Hono();

// Search endpoint proxy
app.get('/api/search/songs', async (c) => {
  const query = c.req.query('query');
  const response = await fetch(`https://www.jiosaavn.com/api.php?__call=autoc
  const data = await response.json();
  return c.json(data);
});

export default app;
```

9. 🗄️ Database Engineering (D1 & KV)

1. Cloudflare D1 (Relational SQLite)

```
-- Users Schema
CREATE TABLE users (
  id TEXT PRIMARY KEY,
  email TEXT UNIQUE NOT NULL,
  name TEXT NOT NULL,
  password_hash TEXT NOT NULL,
  is_admin INTEGER DEFAULT 0,
  created_at TEXT DEFAULT CURRENT_TIMESTAMP
);

-- Liked Songs Relational Table
CREATE TABLE liked_songs (
  id TEXT PRIMARY KEY,
  user_id TEXT NOT NULL,
  song_id TEXT NOT NULL,
  song_data TEXT NOT NULL, -- Stored as JSON string
  liked_at TEXT DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

2. Cloudflare KV (Key-Value Store)

Used for session authorization tokens and real-time user activity heartbeats with automatic **TTL (Time-To-Live)** expiration.

- `session:{token}` → Stores User ID + Email (Expires in 7 days).
- `activity:{userId}` → Stores current playing song + device type (Expires in 5 minutes).

10. 🚀 60 FPS Freezing Fix (Performance Engineering)

> 🔥 **Critical Performance Case Study**
>
> **The Issue:** Browser audio `timeupdate` fired 60 times/second (~60 Hz). Updating React state on every frame triggered 60 global re-renders per second, locking up the main UI thread.

The 4-Part Optimization Solution:

1. **250ms Timestamp Throttling**:

```
// Throttled timeupdate listener in AudioContext:
const now = Date.now();
if (now - lastUpdateRef.current ⩾ 250) {
  setCurrentTime(audioRef.current.currentTime);
  lastUpdateRef.current = now;
}
```

   Result: Reduced global re-renders by >90% (from 60/sec down to 4/sec).

2. **Context Value Memoization (`useMemo`)**: Prevents re-renders of non-consuming child components.
3. **Hidden Component Short-Circuiting**: Added `if (!isOpen) return null;` in `LyricsPanel.jsx` to bypass DOM diffing when hidden.
4. **Handler Stabilization (`useCallback`)**: Stabilizes function references passed to memoized child components.

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11. 🔍 Search Scoring & Ranking Algorithm

Multi-token relevance scoring algorithm for combined queries (e.g. *"Kalyani Shreya Ghoshal"*):

```
function scoreTrack(track, query) {
  const queryTokens = query.toLowerCase().split(/\s+/);
  const title = track.name.toLowerCase();
  const artist = track.artists.primary.map(a => a.name).join(' ').toLowerCase
  const combined = `${title} ${artist}`;

  let score = 0;

  // 1. Exact Title Match Bonus
  if (title === query.toLowerCase()) score += 120;

  // 2. All Tokens Found Across Title + Artist Combined
  const allTokensFound = queryTokens.every(token => combined.includes(token))
  if (allTokensFound) score += 80;

  // 3. Individual Token Score Contributions
  queryTokens.forEach(token => {
    if (title.includes(token)) score += 20;
    if (artist.includes(token)) score += 15;
  });

  // 4. Logarithmic Play Count Boost
  if (track.playCount) {
    score += Math.min(30, Math.log10(Number(track.playCount)) * 5);
  }

  return score;
}
```

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12. 🔐 Security & Cross-Device Sync

- **Password Hashing**: Passwords hashed with bcrypt on server side before D1 storage.
- **Optimistic UI + 8s Sync Loop**: Instant local updates + background polling interval every 8 seconds comparing server timestamps.
- **XSS Protection**: React JSX automatic string escaping + no usage of `dangerouslySetInnerHTML`.

13. 📦 Bundling & Deployment Architecture

- **Cloudflare Pages**: Frontend static asset CDN deployment.
- **Wrangler CLI**: Command line deployment tool (`npx wrangler pages deploy dist`).
- **Git Release Workflow**: Pushed across `main` and `stable-release` branches with version tags (`v4.1.0-stable`).

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14. 🎯 Top 10 Technical Interview Q&As

Q1: Why did you choose Cloudflare Workers over Node.js on AWS?

> "Cloudflare Workers run on V8 JavaScript isolates across 200+ edge locations globally. This gives us sub-5ms latency and zero cold starts, whereas traditional containerized serverless functions suffer 500ms+ cold starts when idle."

Q2: How did you fix performance bottlenecks in your music app?

> "Audio timeupdate events fired 60 times/second, causing 60 re-renders per second. I resolved this by throttling state updates to 250ms intervals, memoizing the Context Provider value with useMemo, and short-circuiting hidden components, reducing CPU overhead by >90%."

Q3: How do you handle password security and authentication?

> "Passwords are hashed with bcrypt before saving to D1 SQL. Authenticated users receive a random session token saved in Cloudflare KV with automated TTL expiration, attached as a Bearer header on protected requests."

Q4: How does cross-device synchronization work?

> "We use an optimistic local-first strategy combined with an 8-second background polling cycle that compares D1 timestamps and updates local state if newer data exists on the server."

Q5: How do you manage gapless track playback?

> "When the current song reaches 90% completion, a preloader instantiates a secondary Audio object to buffer the next stream URL in advance, ensuring instant playback when the ended event fires."

Q6: What is the difference between Cloudflare D1 and KV in your app?

> "D1 is a relational SQLite database used for structured user data, liked songs, and playlists. KV is an in-memory key-value cache with TTL expiration used for session tokens and active user heartbeats."

Q7: Why use Vanilla CSS over Tailwind CSS or CSS-in-JS?

> "Vanilla CSS with Custom Properties provides total control over glassmorphism filters and strict media queries without the JavaScript runtime overhead or bundle bloat of CSS-in-JS."

Q8: How does your search engine handle multi-word queries?

> "We built a multi-token ranking engine that tokenizes queries and scores tracks based on exact title matches, combined artist-title token hits, and logarithmic play count popularity weighting."

Q9: How do you prevent XSS security attacks?

> "React JSX automatically escapes string variables before DOM insertion. We also sanitize user input and avoid dangerous APIs like dangerouslySetInnerHTML."

Q10: How do you ensure the mobile UI works cleanly?

> "We isolated mobile media queries at 768px, transforming the sidebar into a touch-friendly drawer, adjusting player bottom padding to 200px to prevent control overlap, and bypassing Web Audio API filters to conserve mobile battery."

15. 📌 Preparation Checklist

- ✓ Elevator Pitch Memorized
- ✓ 3-Tier Architecture Diagram Clear
- ✓ React 19 State & Context API Mastered
- ✓ Workers vs Lambda Cold Start Ready
- ✓ D1 SQL vs KV Store Differences Clear
- ✓ 250ms Throttle Performance Fix Ready
- ✓ Web Audio API 3-Band EQ Understood
- ✓ LRC Synced Lyrics Regex Parsing Ready
- ✓ Search Scoring Algorithm Code Memorized
- ✓ Top 10 Technical Q&As Practiced